Stochastic Barrier Function Analysis of Market-Relative Wealth via Hybrid Switching Diffusions

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***Abstract*—We analyze the long-run behavior of a portfolio's log-relative wealth process $Z_t = \log(W_t/\mathcal{B}_t)$ under a state-dependent hybrid switching diffusion framework. The portfolio manager selects from three allocation modes — aggressive, moderate, and neutral — switching to recovery strategies with increasing probability as underperformance deepens below a threshold $-L$. Using the barrier function $V(z) = e^{\lambda(-z-L)^+} - 1$, which vanishes on the acceptable performance region $\{z \geq -L\}$ and grows exponentially below it, we prove that $\{Z_t\}$ is bounded from below in probability: $\sup_{t\geq 0} \mathbb{P}(Z_t \leq -R) \to 0$ as $R \to \infty$, with exponential decay rate. The proof combines the generalized Meyer–Ito formula, a Foster–Lyapunov generator condition, Gronwall's inequality, and Markov's inequality. Monte Carlo simulations with $n = 5{,}000$ paths over a horizon of $T = 20$ years confirm the theoretical bounds.**

***Index Terms*—stochastic differential equations, barrier functions, Lyapunov stability, hybrid switching diffusions, portfolio optimization, benchmark-relative performance, Ito calculus, Gronwall inequality**

I. Introduction

A central question in continuous-time portfolio theory is whether an active portfolio manager can ensure that relative performance — measured against a benchmark index — does not deteriorate without bound. While classical results address absolute wealth dynamics, the benchmark-relative setting introduces additional structure: the manager adapts strategy in direct response to underperformance.

This paper formalizes that adaptive response as a *hybrid switching diffusion*. The manager maintains a menu of three portfolio modes, switching among them based on the current log-relative wealth $Z_t = \log(W_t/\mathcal{B}_t)$. When the portfolio falls behind the benchmark by more than L, recovery modes are activated with state-dependent probability, generating upward drift to counteract the slide.

We analyze this system using the barrier function

$$V(z) = e^{\lambda(-z-L)^+} - 1, \quad \lambda > 0, \tag{1}$$

where $(-z-L)^+ = \max(0, -z-L)$. This function is identically zero for $z \geq -L$ and grows exponentially as $z \to -\infty$. It is a Lyapunov function in the classical sense: $V \geq 0$ everywhere and $V(z) = 0$ precisely on the target set $\{z \geq -L\}$, which is the equilibrium set of the controlled system.

Our main result (Theorem **??**) establishes that under mild regularity conditions and a net-stability assumption on the switching policy, $\{Z_t\}_{t\geq 0}$ is bounded from below in probability, uniformly in time. The result is confirmed numerically in Section V.

II. Mathematical Preliminaries

A. Probability Space and Filtration

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$. The filtration $\{\mathcal{F}_t\}_{t\geq 0}$ is an increasing family of σ-algebras ($\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$) encoding the information available at each time t.

B. Conditional Expectation and the Tower Property

Theorem 1 (Tower Property)**.** *For any integrable random variable X and sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, $\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid \mathcal{G}]\big]$.*

C. Brownian Motion and Ito Calculus

Definition 1 (Standard Brownian Motion)**.** *A process $\{B_t\}_{t\geq 0}$ is a standard Brownian motion if: $B_0 = 0$ a.s.; it has independent Gaussian increments $B_t - B_s \sim \mathcal{N}(0, t-s)$; and its paths are continuous a.s.*

Since $\mathbb{E}[B_t \mid \mathcal{F}_s] = B_s$, Brownian motion is a martingale. Its paths are nowhere differentiable, so the quadratic variation $\langle B \rangle_t = t$ is nonzero and the second-order Taylor term survives in any smooth functional expansion. Ito calculus replaces $(dx)^2 = 0$ with $(dB_t)^2 = dt$, $dB_t\, dt = 0$, $(dt)^2 = 0$.

Theorem 2 (Ito's Lemma)**.** *Let X_t satisfy $dX_t = a\, dt + \sigma\, dB_t$. For $f \in C^2$,*

$$df(X_t) = \Big(a f_x + \tfrac{1}{2}\sigma^2 f_{xx}\Big)dt + \sigma f_x\, dB_t.$$

The correction term $\frac{1}{2}\sigma^2 f_{xx}\, dt$ is absent in classical calculus and arises directly from $(dB_t)^2 = dt$.

D. Generalized Ito's Lemma for C^1 Functions

The barrier function (1) is C^∞ on each of the open regions $\{z > -L\}$ and $\{z < -L\}$, but only C^1 at $z = -L$ since the second derivative has a jump there. We therefore invoke the Meyer–Ito formula, which extends Ito's lemma to functions that are C^1 everywhere and C^2 away from a finite set of points.

Theorem 3 (Generalized Ito's Lemma, Meyer–Ito [1]). *Let $f : \mathbb{R} \to \mathbb{R}$ be C^1 everywhere and piecewise C^2 with a kink at c. Let L_t^c denote the local time of X_t at level c. Then*

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)\, dX_s + \frac{1}{2}\int_0^t f''(X_s)\, d\langle X\rangle_s + \frac{1}{2}\Delta f'(c)\, L_t^c,$$

where $\Delta f'(c) = f'(c^+) - f'(c^-)$ is the jump in the first derivative.

E. The Lyapunov Generator

Applying Ito's lemma to a smooth $V(X_t)$ yields $dV = \mathcal{L}V\, dt + \sigma V'\, dB_t$, where the *Lyapunov generator* is $\mathcal{L}V(x) = aV'(x) + \frac{1}{2}\sigma^2 V''(x)$. Taking expectations (the stochastic integral has zero mean): $d\mathbb{E}[V(X_t)]/dt = \mathbb{E}[\mathcal{L}V(X_t)]$. If $\mathcal{L}V(x) < 0$, then V decreases on average — the process is attracted toward the zero set of V.

F. Gronwall's Inequality

Theorem 4 (Gronwall). *Let $m : [0,\infty) \to \mathbb{R}$ be continuous with constants $\alpha > 0$, $\beta \geq 0$ satisfying $m(t) \leq m(0) + \int_0^t (-\alpha\, m(s) + \beta)\, ds$ for all $t \geq 0$. Then*

$$m(t) \leq e^{-\alpha t} m(0) + \frac{\beta}{\alpha}(1 - e^{-\alpha t}), \quad \sup_{t\geq 0} m(t) \leq m(0) + \frac{\beta}{\alpha} < \infty.$$

G. Boundedness from Below in Probability

Definition 2. *A process $\{X_t\}_{t\geq 0}$ is* bounded from below in probability *if for every $\varepsilon > 0$ there exists $R > 0$ such that $\sup_{t\geq 0} \mathbb{P}(X_t \leq -R) \leq \varepsilon$.*

III. Market Model and Problem Formulation

A. Asset Dynamics

The market consists of a risky stock S_t, a benchmark index $\mathcal{B}_t$, and a risk-free asset with constant rate $r > 0$. Both assets follow geometric Brownian motion:

$$\frac{dS_t}{S_t} = \mu_s\, dt + \sigma_s\, dB_t^{(s)}, \qquad \frac{d\mathcal{B}_t}{\mathcal{B}_t} = \mu_m\, dt + \sigma_m\, dB_t^{(m)},$$

with $dB_t^{(s)} dB_t^{(m)} = \rho\, dt$ and constant parameters $\mu_s, \mu_m \in \mathbb{R}$, $\sigma_s, \sigma_m > 0$, $\rho \in [-1, 1]$. We assume continuous trading, self-financing portfolios, and no transaction costs.

B. Portfolio Wealth and Log-Relative Wealth

Let $w(t)$ be the fraction allocated to the risky asset (allowing $w > 1$ for leverage). Under self-financing, $dW_t/W_t = r\, dt + w(t)(\mu_s - r)\, dt + w(t)\sigma_s\, dB_t^{(s)}$. Define $Z_t := \log(W_t/\mathcal{B}_t)$. Applying Ito's lemma to each log and subtracting:

$$dZ_t = a(t)\, dt + \nu(t)\, d\widetilde{B}_t, \tag{2}$$

where $\widetilde{B}_t$ is a standard Brownian motion and

$$a(t) = r + w(t)(\mu_s - r) - \mu_m - \tfrac{1}{2}[w^2(t)\sigma_s^2 - \sigma_m^2], \tag{3}$$

$$\nu^2(t) = w^2(t)\sigma_s^2 + \sigma_m^2 - 2\rho w(t)\sigma_s\sigma_m. \tag{4}$$

When $Z_t > 0$ the portfolio outperforms; when $Z_t < 0$ it underperforms.

C. Three-Mode Switching Policy

The manager selects from three modes partitioned as $\mathcal{Q} = \mathcal{Q}_{\text{rec}} \cup \mathcal{Q}_{\text{neu}}$:

- *Aggressive* ($q = \text{agg}$, $w_{\text{agg}} = 1.5$): leveraged long, maximising upside drift in Z_t.
- *Moderate* ($q = \text{mod}$, $w_{\text{mod}} = 1.0$): fully invested, intermediate drift.
- *Neutral* ($q = \text{neu}$, $w_{\text{neu}} = 0.3$): defensive, low equity exposure.

The recovery set is $\mathcal{Q}_{\text{rec}} = \{\text{agg}, \text{mod}\}$ and the neutral set is $\mathcal{Q}_{\text{neu}} = \{\text{neu}\}$. For threshold $L > 0$, when $Z_t \leq -L$ a recovery mode is activated with probability at least $p(z)$, which increases as $z \to -\infty$. Under this policy the dynamics (2) become the *hybrid switching diffusion*:

$$dZ_t = a_q(Z_t)\, dt + \nu_q(Z_t)\, d\widetilde{B}_t, \quad q = q(Z_t) \in \mathcal{Q}. \tag{5}$$

a) Objective.: Prove that $\sup_{t\geq 0} \mathbb{P}(Z_t \leq -R) \to 0$ as $R \to \infty$.

IV. Barrier Function Analysis

A. Assumptions

We collect the standing hypotheses under which the main results are derived. These assumptions are standard in the theory of hybrid switching diffusions and are satisfied by the three-mode portfolio model of III.

Assumption 1 (Regularity). *For each mode $q \in \mathcal{Q}$, the drift coefficient $a_q : \mathbb{R} \to \mathbb{R}$ and the diffusion coefficient $\nu_q : \mathbb{R} \to \mathbb{R}$ are measurable, locally Lipschitz, and of at-most-linear growth. That is, there exists a constant $C_{\text{lg}} < \infty$ such that*

$$|a_q(z)| + |\nu_q(z)| \leq C_{\text{lg}}(1 + |z|), \quad \forall\, z \in \mathbb{R}.$$

These conditions jointly guarantee the existence of a unique strong solution to the SDE.

Assumption 2 (Adapted Switching). *The mode-selection rule $q(\cdot)$ is $\{\mathcal{F}_t\}$-progressively measurable, so that $q(Z_t)$ is adapted to the filtration generated by the process.*

Assumption 3 (Square Integrability). *For every finite horizon $T < \infty$,*

$$\mathbb{E}\left[\int_0^T \nu_{q(Z_s)}^2(Z_s) s\right] < \infty.$$

This condition ensures that the stochastic integral $M_t = \int_0^t \nu_{q(Z_s)}(Z_s)V'(Z_s)\widetilde{B}_s$ is a true martingale (not merely a local martingale), so that $\mathbb{E}[M_t] = 0$ for all $t \geq 0$.

Assumption 4 (Mode-by-Mode Stability). *There exist constants $\lambda, L, \eta > 0$ and $\kappa < \infty$ such that for all $z \leq -L$:*

- *Recovery modes $q \in \mathcal{Q}_{\text{rec}}$:*

$$-\lambda\, a_q(z) + \tfrac{1}{2}\lambda^2 \nu_q^2(z) \leq -\eta < 0.$$

- *Neutral mode $q \in \mathcal{Q}_{\text{neu}}$:*

$$-\lambda\, a_q(z) + \tfrac{1}{2}\lambda^2 \nu_q^2(z) \leq \kappa < \infty.$$

Assumption 5 (State-Dependent Recovery)**.** *There exists a measurable function $p : (-\infty, -L] \to (0, 1]$ such that*

$$\mathbb{P}\Big(q(Z_t) \in \mathcal{Q}_{\text{rec}} \;\Big|\; Z_t = z\Big) \geq p(z), \quad \forall\, z \leq -L.$$

Intuitively, the deeper the portfolio falls below the threshold, the more likely the manager is to invoke a recovery mode. The function $p(z)$ need not be monotone, but the net stability condition below will require it to be bounded away from zero uniformly.

Assumption 6 (Net Stability)**.** *The effective net recovery rate*

$$\tilde{\eta}(z) := p(z)\,\eta - \big(1 - p(z)\big)\kappa$$

is uniformly positive below the threshold:

$$\tilde{\eta}^* := \inf_{z \leq -L} \tilde{\eta}(z) > 0 \quad \Longleftrightarrow \quad \inf_{z \leq -L} p(z) > \frac{\kappa}{\eta + \kappa}.$$

This is the central stability condition of the paper. It asserts that, on average, the contraction supplied by recovery modes outweighs the potential expansion attributable to the neutral mode. The equivalent form on the right provides a minimum activation frequency*: the manager must engage recovery modes at least a fraction $\kappa/(\eta + \kappa)$ of the time whenever $Z_t \leq -L$.*

B. The Barrier Function and Its Regularity

We define the barrier (Lyapunov) function

$$V(z) := \exp\big(\lambda(-z - L)^+\big) - 1, \quad z \in \mathbb{R}, \tag{6}$$

where $(\cdot)^+ = \max(\cdot, 0)$ and λ is the constant from Assumption 4. We record its basic properties for later use.

(i) *Non-negativity:* $V(z) \geq 0$ for all $z \in \mathbb{R}$, since the exponential is always at least 1.

(ii) *Zero on the target set:* $V(z) = 0$ if and only if $z \geq -L$, because $(-z - L)^+ = 0$ on $\{z \geq -L\}$.

(iii) *Exponential growth in the downside:* For $z < -L$, $V(z) = e^{\lambda(-z-L)} - 1 \to +\infty$ as $z \to -\infty$.

(iv) *Global C^1 regularity:* One computes

$$V'(z) = \begin{cases} 0 & z > -L, \\ -\lambda e^{\lambda(-z-L)} & z < -L. \end{cases}$$

At $z = -L$, the left and right limits both equal 0 from above and $-\lambda$ from below respectively, giving left derivative $-\lambda$ and right derivative 0. However, as a one-sided limit from the left $V'((-L)^-) = -\lambda$ and from the right $V'((-L)^+) = 0$. The function is therefore only $C^1(\mathbb{R})$ in the sense that V is continuous and V' exists everywhere, but V'' has a jump discontinuity at $z = -L$, rendering $V \notin C^2(\mathbb{R})$. This is why the Meyer–Itô formula (Theorem 3) is required in place of the standard Itô lemma.

C. Derivatives and the Itô Inequality

We compute the first and second derivatives of V on each smooth piece explicitly.

Safe region ($z > -L$): Here $(-z - L)^+ = 0$, so $V(z) = 0$ identically, giving

$$V'(z) = 0, \qquad V''(z) = 0.$$

Downside region ($z < -L$): Here $(-z - L)^+ = -z - L > 0$, so $V(z) = e^{\lambda(-z-L)} - 1$. Differentiating:

$$V'(z) = -\lambda e^{\lambda(-z-L)},$$
$$V''(z) = \lambda^2 e^{\lambda(-z-L)}.$$

Note the useful identity $e^{\lambda(-z-L)} = V(z) + 1$, which will allow us to express the generator in terms of V itself.

Jump in the first derivative at $z = -L$:

$$\Delta V'(-L) := V'((-L)^+) - V'((-L)^-) = 0 - (-\lambda) = \lambda > 0.$$

Since $\Delta V'(-L) > 0$, the Meyer–Itô formula contributes a local-time correction term $\frac{1}{2}\lambda L_t^{-L} \geq 0$ at the kink.

We now apply Theorem 3 to $V(Z_t)$.

Lemma 1 (Itô Inequality)**.** *Define the stochastic integral $M_t := \int_0^t \nu_{q(Z_s)}(Z_s)\, V'(Z_s)\widetilde{B}_s$. Under Assumption 3, M_t is a martingale with $\mathbb{E}[M_t] = 0$. The following inequality holds pathwise for all $t \geq 0$:*

$$V(Z_t) \leq V(Z_0) + \int_0^t \mathcal{L}_{q(Z_s)} V(Z_s) s + M_t, \tag{7}$$

where the generator acting on V at mode q is $\mathcal{L}_q V(z) := a_q(z)\, V'(z) + \frac{1}{2}\nu_q^2(z)\, V''(z)$.

Proof. By the Meyer–Itô formula (Theorem 3) applied to the C^1 function V with kink at $c = -L$:

$$V(Z_t) = V(Z_0) + \int_0^t V'(Z_s) Z_s + \frac{1}{2}\int_0^t V''(Z_s)\langle Z\rangle_s + \frac{1}{2}\,\Delta V'(-L)\, L_t^{-L}. \tag{8}$$

Substituting $Z_s = a_{q(Z_s)}(Z_s)s + \nu_{q(Z_s)}(Z_s)\widetilde{B}_s$ and $\langle Z\rangle_s = \nu^2_{q(Z_s)}(Z_s)s$, equation (8) becomes

$$V(Z_t) = V(Z_0) + \int_0^t \left[a_{q(Z_s)}(Z_s)V'(Z_s) + \frac{1}{2}\nu^2_{q(Z_s)}(Z_s)V''(Z_s)\right]s + \int_0^t \nu_{q(Z_s)}(Z_s)V'(Z_s)\widetilde{B}_s + \frac{\lambda}{2}\, L_t^{-L}.$$

Recognising the integrand as $\mathcal{L}_{q(Z_s)}V(Z_s)$ and M_t as the stochastic integral, and using the fact that $L_t^{-L} \geq 0$ to drop the local-time term, we arrive at the inequality (7). □

D. Generator Computation

We now evaluate $\mathcal{L}_q V(z) = a_q(z)V'(z) + \frac{1}{2}\nu_q^2(z)V''(z)$ on each region explicitly.

Safe region ($z > -L$): Since $V'(z) = V''(z) = 0$,

$$\mathcal{L}_q V(z) = 0.$$

The barrier is completely inert above the threshold: no matter what mode is active or what the drift and volatility are, the generator vanishes. This is an important structural feature of the chosen barrier: it imposes no "cost" on normal fluctuations above $-L$.

Downside region ($z < -L$): Substituting $V'(z) = -\lambda e^{\lambda(-z-L)}$ and $V''(z) = \lambda^2 e^{\lambda(-z-L)}$,

$$\begin{aligned}\mathcal{L}_q V(z) &= a_q(z)\big(-\lambda e^{\lambda(-z-L)}\big) + \frac{1}{2}\nu_q^2(z)\big(\lambda^2 e^{\lambda(-z-L)}\big)\\ &= e^{\lambda(-z-L)}\left[-\lambda a_q(z) + \frac{1}{2}\lambda^2\nu_q^2(z)\right].\end{aligned}$$

Using the identity $e^{\lambda(-z-L)} = V(z) + 1$, we obtain the factored form

$$\mathcal{L}_q V(z) = \big(V(z)+1\big)\left[-\lambda a_q(z) + \frac{1}{2}\lambda^2\nu_q^2(z)\right], \quad z < -L. \tag{9}$$

The factor $(V(z) + 1) > 0$ is always positive, so the sign of $\mathcal{L}_q V(z)$ is entirely determined by the bracket $\Phi_q(z) := -\lambda a_q(z) + \frac{1}{2}\lambda^2\nu_q^2(z)$, which is exactly what Assumption 4 controls.

E. Foster–Lyapunov Drift Condition

We now leverage the mode-by-mode stability (Assumption 4) and the state-dependent recovery probability (Assumption 5) to derive a global, unconditional drift inequality on V.

Lemma 2 (Conditional Generator Bound)**.** *For all $z \le -L$,*

$$\mathbb{E}\Big[\mathcal{L}_{q(Z_t)}V(Z_t) \;\Big|\; Z_t = z\Big] \le -\tilde{\eta}^*\big(V(z)+1\big).$$

Proof. We treat the boundary and interior separately.

Boundary ($z = -L$): At $z = -L$ we have $V(-L) = 0$, and from the safe-region formula $\mathcal{L}_q V(-L) = 0$ for all modes q. The right-hand side at $z = -L$ is $-\tilde{\eta}^*(0 + 1) = -\tilde{\eta}^* < 0$. Hence the inequality $0 \le -\tilde{\eta}^*$ is satisfied trivially since the left side is 0 and $-\tilde{\eta}^* < 0$.

Interior ($z < -L$): Using the factored generator (9) and conditioning on the mode selection, we write

$$\mathbb{E}\Big[\mathcal{L}_{q(Z_t)}V(Z_t) \;\Big|\; Z_t = z\Big] = \big(V(z)+1\big)\,\mathbb{E}\Big[\Phi_{q(Z_t)}(z) \;\Big|\; Z_t = z\Big]$$

We decompose the expectation of $\Phi_{q(Z_t)}(z)$ over the two mode classes. Let $p_z := \mathbb{P}(q(Z_t) \in \mathcal{Q}_{\mathrm{rec}} \mid Z_t = z) \ge p(z)$ and $1 - p_z \le 1 - p(z)$. By Assumption 4:

$$\begin{aligned}\mathbb{E}\Big[\Phi_{q(Z_t)}(z) \;\Big|\; Z_t = z\Big] &= p_z\,\mathbb{E}\Big[\Phi_q(z) \;\Big|\; Z_t = z,\, q \in \mathcal{Q}_{\mathrm{rec}}\Big]\\ &\quad + (1-p_z)\,\mathbb{E}\Big[\Phi_q(z) \;\Big|\; Z_t = z,\, q \in \mathcal{Q}_{\mathrm{neu}}\Big]\\ &\le p_z \cdot (-\eta) + (1-p_z)\cdot\kappa\\ &= -p_z\,\eta + (1-p_z)\,\kappa\\ &= -\big[p_z\,\eta - (1-p_z)\,\kappa\big]\\ &\le -\tilde{\eta}(z) \le -\tilde{\eta}^*,\end{aligned}$$

where the penultimate inequality uses $p_z \ge p(z)$ together with the fact that $\tilde{\eta}(z) = p(z)\eta - (1-p(z))\kappa$ is increasing in $p(z)$, and the final step applies the infimum from Assumption 6. Since $(V(z) + 1) > 0$, multiplying through gives

$$\mathbb{E}\Big[\mathcal{L}_{q(Z_t)}V(Z_t) \;\Big|\; Z_t = z\Big] \le (V(z)+1)\cdot(-\tilde{\eta}^*) = -\tilde{\eta}^*\big(V(z)+1\big),$$

as claimed. □

Lemma 3 (Global Foster–Lyapunov Condition)**.** *There exists a finite constant $d < \infty$ such that for all $z \in \mathbb{R}$,*

$$\mathbb{E}\Big[\mathcal{L}_{q(Z_t)}V(Z_t) \;\Big|\; Z_t = z\Big] \le -\tilde{\eta}^*\,V(z) + d. \tag{10}$$

Proof. We establish (10) separately on each region.

Safe region ($z > -L$): Here $V(z) = 0$ and $\mathcal{L}_q V(z) = 0$ for all q, so the left-hand side of (10) is 0. The right-hand side is $-\tilde{\eta}^* \cdot 0 + d = d \ge 0$ for any $d \ge 0$. Hence (10) holds trivially on the safe region.

Downside region ($z \le -L$): Lemma 2 gives

$$\mathbb{E}\Big[\mathcal{L}_{q(Z_t)}V(Z_t) \;\Big|\; Z_t = z\Big] \le -\tilde{\eta}^*\big(V(z)+1\big) = -\tilde{\eta}^*V(z) - \tilde{\eta}^*.$$

Since $-\tilde{\eta}^* < 0$, this is even stronger than (10) with $d = \tilde{\eta}^*$, i.e. the bound $-\tilde{\eta}^*V(z) - \tilde{\eta}^* \le -\tilde{\eta}^*V(z) + d$ holds whenever $d \ge \tilde{\eta}^*$.

Conclusion: Choosing $d = \tilde{\eta}^* < \infty$, inequality (10) holds for all $z \in \mathbb{R}$. □

Remark 1. *The condition* (10) *is a classical* Foster–Lyapunov *or* negative drift outside a petite set *condition. Here the "petite set" is $\{z \ge -L\}$, on which V vanishes identically, and d quantifies the maximum rate at which the barrier can grow near the boundary.*

F. Main Results

We now assemble the preceding lemmas to derive the uniform moment bound on V and the main tail probability estimate.

Theorem 5 (Uniform Barrier Moment Bound)**.** *Under Assumptions 1–6,*

$$\sup_{t\ge 0}\ \mathbb{E}\Big[e^{\lambda(-Z_t-L)^+} - 1\Big] \;\le\; C \;:=\; V(Z_0) + \frac{d}{\tilde{\eta}^*} \;<\; \infty.$$

Proof. The proof proceeds in four steps.

Step 1: From the Itô inequality to an integral inequality for the expected barrier.

Take expectations on both sides of the Itô inequality (7):

$$\mathbb{E}[V(Z_t)] \le V(Z_0) + \mathbb{E}\left[\int_0^t \mathcal{L}_{q(Z_s)} V(Z_s) s\right] + \mathbb{E}[M_t].$$

By Assumption 3, M_t is a square-integrable martingale, so $\mathbb{E}[M_t] = 0$. Applying Fubini's theorem to interchange the expectation and the time integral (justified by the linear growth bound in Assumption 1 and integrability of V on finite intervals):

$$\mathbb{E}[V(Z_t)] \le V(Z_0) + \int_0^t \mathbb{E}\big[\mathcal{L}_{q(Z_s)} V(Z_s)\big]\, s. \tag{11}$$

Step 2: Apply the Foster–Lyapunov condition inside the integral.

For each fixed $s \ge 0$, apply the Tower Property (Theorem 1) to the conditional generator bound (10):

$$\begin{aligned}\mathbb{E}\big[\mathcal{L}_{q(Z_s)} V(Z_s)\big] &= \mathbb{E}\Big[\mathbb{E}\Big[\mathcal{L}_{q(Z_s)} V(Z_s) \;\Big|\; Z_s\Big]\Big] \\ &\le \mathbb{E}[-\tilde{\eta}^* V(Z_s) + d] \\ &= -\tilde{\eta}^* \,\mathbb{E}[V(Z_s)] + d.\end{aligned}$$

Substituting into (11):

$$\mathbb{E}[V(Z_t)] \le V(Z_0) + \int_0^t \left(-\tilde{\eta}^* \,\mathbb{E}[V(Z_s)] + d\right) s. \tag{12}$$

Step 3: Apply Gronwall's inequality.

Set $m(t) := \mathbb{E}[V(Z_t)]$, $\alpha := \tilde{\eta}^* > 0$, and $\beta := d \ge 0$. Then (12) reads

$$m(t) \le m(0) + \int_0^t \big(-\alpha\, m(s) + \beta\big) s,$$

which is precisely the hypothesis of Gronwall's inequality (Theorem 4). Applying it yields

$$m(t) \le e^{-\alpha t}\, m(0) + \frac{\beta}{\alpha}\big(1 - e^{-\alpha t}\big)\,.$$

Step 4: Take the supremum over time.

Since $e^{-\alpha t} \le 1$ and $(1 - e^{-\alpha t}) \le 1$ for all $t \ge 0$,

$$\begin{aligned}\sup_{t\ge 0} m(t) &\le \sup_{t\ge 0}\left[e^{-\alpha t}\, m(0) + \frac{\beta}{\alpha}\big(1 - e^{-\alpha t}\big)\right] \\ &\le m(0) + \frac{\beta}{\alpha} \\ &= V(Z_0) + \frac{d}{\tilde{\eta}^*} =: C < \infty.\end{aligned}$$

This completes the proof. □

Theorem 6 (Boundedness from Below in Probability)**.** *Under Assumptions 1–6, for all $R > L$,*

$$\sup_{t\ge 0} \mathbb{P}(Z_t \le -R) \;\le\; \frac{C}{e^{\lambda(R-L)} - 1} \;\longrightarrow\; 0 \quad \textit{as } R \to \infty. \tag{13}$$

Proof. The argument combines the moment bound of Theorem 5 with a level-set lower bound on V and Markov's inequality.

Step 1: Lower bound on V on the event $\{Z_t \le -R\}$.

Fix $R > L$ and $t \ge 0$. On the event $\{Z_t \le -R\}$, we have $-Z_t - L \ge R - L > 0$, so $(-Z_t - L)^+ = -Z_t - L \ge R - L$. Therefore,

$$V(Z_t) = e^{\lambda(-Z_t - L)^+} - 1 \ge e^{\lambda(R-L)} - 1 > 0.$$

That is, the barrier V is bounded below by the positive constant $e^{\lambda(R-L)} - 1$ whenever Z_t is at or below level $-R$.

Step 2: Apply Markov's inequality.

From Step 1, for any $t \ge 0$:

$$\mathbb{E}[V(Z_t)] \ge \mathbb{E}\big[V(Z_t)\,\mathbf{1}_{\{Z_t \le -R\}}\big] \ge \big(e^{\lambda(R-L)} - 1\big)\,\mathbb{P}(Z_t \le -R).$$

Rearranging:

$$\mathbb{P}(Z_t \le -R) \le \frac{\mathbb{E}[V(Z_t)]}{e^{\lambda(R-L)} - 1}.$$

Step 3: Apply the uniform moment bound.

Since Theorem 5 gives $\mathbb{E}[V(Z_t)] \le C$ for all $t \ge 0$,

$$\mathbb{P}(Z_t \le -R) \le \frac{C}{e^{\lambda(R-L)} - 1} \quad \text{for every } t \ge 0.$$

Taking the supremum over t on the left side preserves the inequality, yielding (13).

Step 4: Convergence to zero.

As $R \to \infty$, the denominator $e^{\lambda(R-L)} - 1 \to +\infty$ while the numerator $C < \infty$ is fixed. Hence the right-hand side of (13) converges to 0. Since Definition **??** requires exactly this for every $\varepsilon > 0$ (take $R^* = L + \lambda^{-1}\log(C/\varepsilon + 1)$), the process $\{Z_t\}$ is bounded from below in probability. □

Corollary 1. *For large R,*

$$\frac{C}{e^{\lambda(R-L)} - 1} \sim C\, e^{-\lambda(R-L)},$$

so the tail probability decays exponentially *in R at rate λ. Equivalently,* $\sup_{t\ge 0} \mathbb{P}(Z_t \le -R) = O(e^{-\lambda R})$ *as $R \to \infty$. The decay rate λ is the same parameter appearing in the mode-by-mode stability Assumption 4.*

V. Numerical Simulation

We verify the theoretical results via Euler–Maruyama Monte Carlo simulation.

A. Parameter Specification

Table I summarises all parameters. The market parameters are standard continuous-time finance defaults. The neutral mode has negative drift $a_{\text{neu}} = -0.0196$ because $w_{\text{neu}} = 0.3$ is insufficient to overcome the benchmark's excess return; this makes Assumption 4 non-trivial, and the net stability condition (Assumption 6) is the binding constraint.

TABLE I
SIMULATION PARAMETERS

Parameter	Symbol	Value
Stock drift / vol	$\mu_s,\ \sigma_s$	0.10, 0.20
Benchmark drift / vol	$\mu_m,\ \sigma_m$	0.08, 0.15
Correlation / risk-free	$\rho,\ r$	0.70, 0.03
Aggressive allocation	w_{agg}	1.5 (leveraged)
Moderate allocation	w_{mod}	1.0 (fully invested)
Neutral allocation	w_{neu}	0.3 (defensive)
Threshold / barrier exp.	$L,\ \lambda$	1.0, 0.075
Recovery prob. base	$p_{\min}$	0.70
Computed $a_q,\ \nu_q$	$a_{\text{agg}}, a_{\text{mod}}, a_{\text{neu}}$	0.0213, 0.0113, −0.0196
	$\nu_{\text{agg}}, \nu_{\text{mod}}, \nu_{\text{neu}}$	0.2225, 0.1432, 0.1162
$\eta,\ \kappa,\ \tilde{\eta}^*$	—	$7.86\times10^{-4},\ 1.50\times10^{-3},\ 9.9\times10^{-5}$
Bound	C	1.000
Paths / steps / horizon	$n,\ N,\ T$	5000, 20000, 20 yr

The recovery probability is $p(z) = \min(1,\ 0.70 + 0.30\,(-z - L)^+)$, reaching $p_{\min} = 0.70$ at the boundary $z = -L$ and ramping toward 1 as the portfolio falls deeper. One verifies $p_{\min} = 0.70 > \kappa/(\eta + \kappa) \approx 0.657$, so Assumption 6 holds with $\tilde{\eta}^* = 9.9 \times 10^{-5} > 0$. Within the recovery region, the probability of activating the aggressive mode ramps from 0.50 toward 0.70, while the moderate mode fills the remainder.

B. Sample Paths

Figure 1 shows eight representative paths over $T = 20$ years. All paths fluctuate freely above the threshold $-L = -1.0$ and periodically dip into the recovery region (shaded red). Crucially, no path drifts arbitrarily downward: each excursion below the threshold activates recovery modes with increasing probability, generating the visible upward corrections. Across all 5,000 paths, the process spends approximately 2.4% of total time in the recovery region.

C. Tail Probability Decay

Figure 2 plots $\sup_t \mathbb{P}(Z_t \le -R)$ against the theoretical bound $C(e^{\lambda(R-L)}-1)^{-1}$ on a logarithmic scale. The simulated tail probability lies strictly below the theoretical envelope at every $R > L$, confirming Theorem 6. Both curves decay at the same exponential rate $\lambda = 0.075$ for large R, visible as equal slopes on the log-scale plot. Table II tabulates selected values; the large gap between simulation and bound quantifies the conservatism of the Foster–Lyapunov estimate.

TABLE II
SIMULATED VS. THEORETICAL TAIL PROBABILITIES

R	Simulated $\sup_t \mathbb{P}(Z_t \le -R)$	Bound $C/(e^{\lambda(R-L)} - 1)$
1.5	0.01140	26.170
2.0	0.00140	12.840
2.5	0.00020	8.398
3.0	< 0.0001	6.179

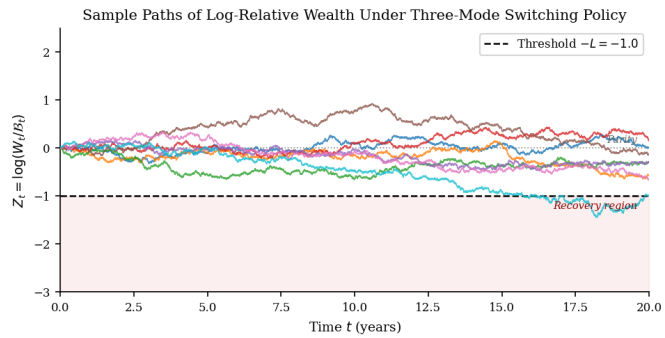

Fig. 1. Eight sample paths of $Z_t = \log(W_t/\mathcal{B}_t)$ under the three-mode switching policy. The dashed line marks the threshold $-L = -1.0$; the red-shaded region is where the policy activates recovery modes. No path drifts to $-\infty$.

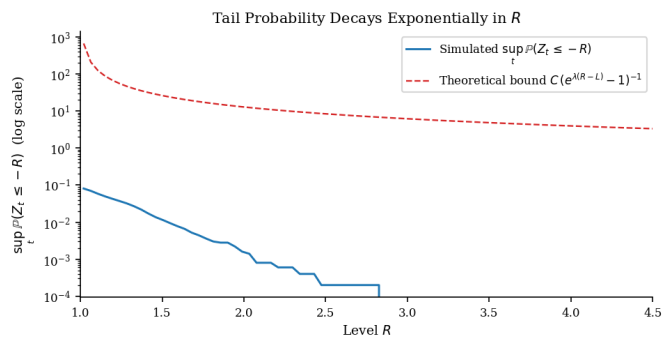

Fig. 2. Simulated supremum tail probability (solid) and theoretical bound (dashed) on a log scale. Both decay exponentially at rate $\lambda = 0.075$; the bound is valid at every $R > L = 1.0$.

D. Stationarity of the Distribution

Figure 3 shows the empirical distribution of Z_t at $t = 2, 8, 20$ years. After a short initial transient, the distribution stabilises and the histograms at $t = 8$ and $t = 20$ are nearly identical, suggesting convergence to a stationary distribution. No probability mass accumulates near $-\infty$ across

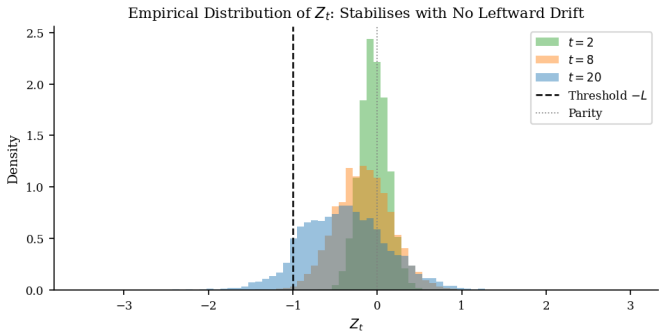

Fig. 3. Empirical distribution of Z_t at $t = 2, 8, 20$ years. The distribution stabilises rapidly; the left tail remains bounded with no drift toward $-\infty$, consistent with boundedness from below in probability.

any snapshot, which is precisely the content of Theorem 6. The distribution is positively skewed: the switching policy pushes Z_t upward from the downside region, but no corresponding mechanism caps the upside, allowing the right tail to spread freely over time.

VI. Financial Interpretation

Theorem 6 guarantees that the log-relative wealth $Z_t = \log(W_t/\mathcal{B}_t)$ cannot drift arbitrarily far below the benchmark with significant probability, regardless of the time horizon. For any tolerance $\varepsilon > 0$ there exists a floor $-R^*$ such that $\sup_{t\geq 0} \mathbb{P}(Z_t \leq -R^*) \leq \varepsilon$.

The barrier $V(z) = e^{\lambda(-z-L)^+} - 1$ formalises the manager's risk objective: it assigns zero cost to any $z \geq -L$ and exponentially growing cost to deeper underperformance. The generator analysis makes the mechanics precise: above the threshold $\mathcal{L}_q V = 0$, so no corrective action is needed, and the barrier is completely insensitive to normal market fluctuations. Only when $Z_t < -L$ does the negative drift condition engage, and the three-mode switching policy ensures that recovery dominates on average.

The three-mode structure captures a realistic graduated response: as underperformance deepens from moderate (z near $-L$) to severe ($z \ll -L$), the probability of activating the aggressive leveraged mode increases, providing stronger corrective drift. The net stability condition (Assumption **??**) quantifies the minimum aggressiveness required: $p_{\min} > \kappa/(\eta + \kappa) \approx 0.657$. This is a hard lower bound on how often the manager must use recovery modes when below the threshold — a practically interpretable constraint that can be incorporated directly into a portfolio mandate.

VII. Conclusion

We have established that the log-relative wealth of a portfolio under a three-mode state-dependent switching policy is bounded from below in probability, with tail bound $\sup_{t\geq 0} \mathbb{P}(Z_t \leq -R) \leq C(e^{\lambda(R-L)} - 1)^{-1}$. The proof proceeds in five steps: (1) choose the barrier $V(z) = e^{\lambda(-z-L)^+} - 1$, vanishing on the target set $\{z \geq -L\}$; (2) apply the Meyer–Ito formula and drop the non-negative local time term; (3) establish the global Foster–Lyapunov condition using mode-by-mode stability (A4), state-dependent recovery probability (A5), and net stability (A6); (4) apply Gronwall's inequality to obtain $\sup_{t\geq 0} \mathbb{E}[V(Z_t)] \leq C$; (5) apply Markov's inequality for the exponential tail bound. Monte Carlo simulation with 5,000 paths over 20 years confirms all theoretical predictions: the barrier expectation remains uniformly bounded well below C, the tail probability decays exponentially in R, and the distribution stabilises with no leftward drift.

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